APOLLO

Hand-Calculation Outline

and Worked Answer Key



Comic-book mission montage / AI-generated ink illustration / not to scale.

Mission Control copy — keep answers covered

Every mandatory calculation for the Apollo 11 beginner game and Apollo 13 advanced game: inputs, equations, slide-rule procedures, correct answers, decision gates, error consequences, and read-aloud narration examples for Mission Control.

10 Apollo 11 checkpoints

10 Apollo 13 checkpoints

Companion player book: `apollo-game.pdf`

Use guard digits to grade; expect about three significant figures from a player. Do not demand calculator precision from a slide rule.

Historical schedules are cited. Invented telemetry and thresholds are educational models, not NASA operational limits.

Referee protocol and a worked error checkpoint

Read the player book's full rules first. Apollo 11 starts with command reserve 6; Apollo 13 starts with 4. Each card is graded on the *first committed sheet*; charge only its worst band once. GREEN costs 0, AMBER 1, RED 2. Continue at zero, stop below zero.

Numerical bands

Ordinary outputs: $\epsilon = 100|q - q_{\text{ref}}|/|q_{\text{ref}}|$. GREEN $\leq 1\%$, AMBER $> 1\%$ through 5%, RED $> 5\%$. Clock-format GET / intervals: GREEN within 1 s, AMBER through 60 s, RED beyond. Computed decimal durations use relative bands. Angles on 11-10, 13-08, 13-09: GREEN within $0.05°$, AMBER through $0.20°$, RED beyond. Wrong signs/quadrants, missing outputs, missing/incompatible units, failed physical gates, or wrong discrete choices are RED.

For physical gates, evaluate unrounded quantities. Do not mark a correct rejected-profile negative remainder wrong merely because it is negative. Only the *selected* plan must meet the stated constraint. The player must use the slide rule for the required products, quotients, roots, and logarithms and record the work.

Example: card 11-06. Crew commits fuel endurance 82.0 s, descent time 60.0 s, and chooses the correct 20 s diversion. Reference endurance is 85.7142857 s;

$$\epsilon = 100\,|82.0 - 85.7142857|/85.7142857 = 4.33333\%.$$

If all other outputs are GREEN, the checkpoint is **AMBER**. Charge 1 reserve, narrate the less desirable landing site, and require the corrected 85.7 s endurance / 25.7 s hover margin.

If the crew instead selects a 30 s diversion, the physical gate fails: it needs $0.700(60 + 30) = 63.0$ kg but has only 60.0. Grade **RED**, charge 2 once (not once per output), and narrate the wave-off rehearsal. If reserve remains nonnegative, correct the sheet and resume at the nominal landing baseline. If negative, stop with the Apollo 11 safe lunar-orbit abort ending.

State discipline. Reserve and the Apollo 11 EVA narrative flag are the only persistent error effects. Do not also consume invented minutes, water, or fuel because of a warning. All accepted pads must be corrected to GREEN quality before continuing. Game failure is a simulation result, not a claim that the historical crew failed.

Read-aloud help. The narration appendix starts on page 22. It includes launch and coast transitions, briefings and GREEN / AMBER / RED examples for every card, the cartridge-adapter scene, and all mission endings. Read briefings before commitment; keep outcome scripts hidden until grading.

11-01 Commit to the Moon

Timeline / provenance: Planned TLI ignition 002:44:26; 320 s burn. [F1, Table 1-1]

Calculation outline — all required outputs

Convert the planned 10,451.2 ft/s velocity change to m/s. Find its average acceleration over 5 min 20 s. Compute the cutoff GET.

Hand method and slide-rule scales. Convert minutes by hand. On C/D multiply velocity by 0.3048, then divide by elapsed seconds: $\bar{a} = \Delta v/t$. Add the duration to GET on paper. This is a whole-burn average, not constant actual thrust acceleration.

Worked answers (guard digits are not player precision)

$\Delta v = 10{,}451.2(0.3048) = 3185.52576$ m/s, so report $\boxed{3.19 \times 10^3 \text{ m/s}}$. $t = 5(60) + 20 = 320$ s; $\bar{a} = 3185.52576/320 = \boxed{9.95 \text{ m/s}^2}$. Cutoff 002:44:26 + 00:05:20 = $\boxed{002:49:46}$. An estimate near 10^4 ft/s and 3×10^3 m/s catches a factor-ten error.

Physical / decision gate. Write m/s, m/s^2, and h:m:s on the pad. Missing or incompatible units are RED. The modeled burn is commanded in the forward direction.

Checkpoint consequences. AMBER: guidance flags a targeting discrepancy; charge 1 reserve for a corrective update. RED: the departure targeting test fails; charge 2 for a simulator abort-and-replan. A depleted reserve here means a safe Earth-orbit abort, not a lunar landing.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-02 Clear the spent stage

Timeline / provenance: Planned evasive maneuver 004:39:44.9; 2.8 s. [F1, Table 1-1]

Calculation outline — all required outputs

For $(\Delta v_x, \Delta v_y, \Delta v_z) = (5.1, 0, 19.0)$ ft/s, find total velocity change in ft/s and m/s. A training model supplies average acceleration 2.14 m/s^2; find burn duration.

Hand method and slide-rule scales. Square on A (or multiply C/D), add the squares with pencil, take the square root on A/D: $\Delta v = \sqrt{\Delta v_x^2 + \Delta v_y^2 + \Delta v_z^2}$. Convert on C/D and divide by $\bar{a}$.

Worked answers (guard digits are not player precision)

$5.1^2 + 19^2 = 26.01 + 361 = 387.01$. $\Delta v = \sqrt{387.01} = 19.67257$ ft/s $\rightarrow$ $\boxed{19.7 \text{ ft/s}}$. $\Delta v = 19.67257(0.3048) = \boxed{6.00 \text{ m/s}}$. $t = 5.996199/2.14 = \boxed{2.80 \text{ s}}$, matching the planned 2.8 s to slide-rule precision. Use the table's vector / required-Δv data here, not the different 15.6 ft/s value in its ΔV_C column.

Physical / decision gate. Use the vector magnitude, not $5.1 + 19.0$. Preserve the supplied component directions on the command pad. The 2.14 m/s^2 acceleration is invented training telemetry.

Checkpoint consequences. AMBER: separation is slower or faster than expected; charge 1 for a tracking update. RED: the safe-separation prediction fails; charge 2 for an emergency avoidance rehearsal before continuing.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-03 Capture lunar orbit

Timeline / provenance: Planned LOI1 ignition 075:54:28.4; 358.9 s. [F1, Table 1-1]

Calculation outline — all required outputs

Use components $(-2891.8, -433.1, +20.4)$ ft/s to find Δv in ft/s and m/s. Find average acceleration for 5 min 58.9 s. Compute cutoff GET.

Hand method and slide-rule scales. Use A/D for squares and roots, pencil for addition, C/D for unit conversion and division. Use $\bar{a} = \Delta v/t$. Clock arithmetic is sexagesimal; keep the tenths of a second.

Worked answers (guard digits are not player precision)

$\Sigma\Delta v_i^2 = 8{,}550{,}499.01$ (ft/s)^2. Square root $= 2924.12363$ ft/s $\rightarrow$ $\boxed{2.92 \times 10^3 \text{ ft/s}}$. Multiply by 0.3048 to get 891.273 m/s $\rightarrow$ $\boxed{891 \text{ m/s}}$. $t = 358.9$ s and $\bar{a} = 891.273/358.9 = \boxed{2.48 \text{ m/s}^2}$. Cutoff 075:54:28.4 + 00:05:58.9 = $\boxed{076{:}00{:}27.3}$. The table rounds required Δv to 2924.1 ft/s; its separate $\Delta V_C = 2914.8$ ft/s is not substituted.

Physical / decision gate. Keep all three component signs. A correct magnitude with reversed retrograde direction is RED. This scalar exercise does not compute an orbit or prove capture in the real world.

Checkpoint consequences. AMBER: tracking predicts an off-target lunar orbit; charge 1 for a trim update. RED: the simulator rejects capture; charge 2 for a contingency return rehearsal. Reserve exhaustion means abort the landing and use the game's safe-return ending.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-04 Work the lunar clock

Timeline / provenance: Lunar-orbit operations between LOI2 and LM separation. [F1, p. 1-3]

Calculation outline — all required outputs

An average orbit lasts 118.2 min, including 47 min of night. Find daylight minutes, average angular rate in degrees/min, and the percentage of an orbit with MSFN coverage if coverage is 72 min. Is a 65 min daylight task feasible in one daylight interval?

Hand method and slide-rule scales. Subtract with pencil, then use C/D for 360/118.2 and 100(72/118.2). Compare the task with daylight duration. Use the supplied full period, not the plan's rounded 3°/min.

Worked answers (guard digits are not player precision)

Daylight = 118.2 − 47 = $\boxed{71.2 \text{ min}}$. Angular rate = 360/118.2 = $\boxed{3.05^\circ/\text{min}}$. Coverage = 100(72/118.2) = $\boxed{60.9\%}$. $\boxed{\text{GO}}$ for the daylight task: 71.2 − 65 = $\boxed{6.20 \text{ min}}$ spare. The plan gives approximately 3°/min; using the more precise period explains the small difference. There is insufficient information to require continuous radio contact throughout the task.

Physical / decision gate. The 65 min task must fit daylight. Do not treat 72 min of coverage as 72 min of daylight, or claim that their overlap is known. The task duration is a training scenario.

Checkpoint consequences. AMBER: a mistimed observation loses useful coverage; charge 1 for rescheduling. RED: the crew confuses the two windows and misses the handover; charge 2 for a complete scheduling rebuild.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-05 Lower Eagle's perilune

Timeline / provenance: Planned DOI 101:38:48, 28.5 s; PDI 102:35:13. [F1, Table 1-2]

Calculation outline — all required outputs

Use the table's required $\Delta v = 70.0$ ft/s. Convert it to m/s; compute average acceleration over 28.5 s. Find DOI cutoff GET and the coast interval from cutoff to PDI.

Hand method and slide-rule scales. C/D handles conversion and $\Delta v/t$. Add the burn on paper, then subtract cutoff from PDI, borrowing 60 seconds or minutes when necessary.

Worked answers (guard digits are not player precision)

$70(0.3048) = \boxed{21.3 \text{ m/s}}$. $\bar{a} = 21.336/28.5 = \boxed{0.749 \text{ m/s}^2}$. Cutoff 101:38:48 + 00:00:28.5 = $\boxed{101:39:16.5}$. PDI minus cutoff: 102:35:13 − 101:39:16.5 = $\boxed{00:55:56.5}$. The printed components 67.46, −28.68, −12.51 ft/s would give about 74.4 ft/s, so the specified 70.0 ft/s input is deliberate.

Physical / decision gate. Use required Δv, not a reconstruction from the printed components: this table's DOI components do not exactly reproduce 70. The game does not resolve that source discrepancy.

Checkpoint consequences. AMBER: the coast briefing is late; charge 1 for an expedited pad review. RED: a descent-readiness check is missed; charge 2 for a full rehearsal. If reserve is exhausted, cancel PDI and rendezvous with Columbia in the game's safe-return ending.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-06 Land or wave off

Timeline / provenance: Planned PDI 102:35:13 to touchdown 102:47:11. [F1, pp. 1-9, 3-69]

Calculation outline — all required outputs

First find touchdown GET and average whole-descent acceleration from $\Delta v = 6766$ ft/s and 11 min 58 s. At the terminal gate, altitude is 120 m and downward speed is held at 2.00 m/s. Fuel is 60.0 kg and consumption is 0.700 kg/s. Find descent time, fuel endurance, and maximum extra hover time. Choose the longest safe diversion from 10, 20, or 30 s; find fuel left.

Hand method and slide-rule scales. Use C/D to convert velocity, divide for average acceleration, divide altitude by speed, and divide fuel by flow. Subtract endurance minus descent time. For a diversion d, $m_{\text{left}} = 60 - 0.700(120/2 + d)$.

Worked answers (guard digits are not player precision)

Touchdown = $\boxed{102:47:11}$. $t = 718$ s; $\Delta v = 6766(0.3048) = 2062.2768$ m/s; $\bar{a} = \boxed{2.87\ \text{m/s}^2}$. Terminal descent $= 120/2 = \boxed{60.0\ \text{s}}$. Endurance $= 60/0.700 = \boxed{85.7\ \text{s}}$; extra hover $= 85.7143 - 60 = \boxed{25.7\ \text{s}}$. Choose $\boxed{20\ \text{s}}$. Fuel left $= 60 - 0.700(80) = \boxed{4.00\ \text{kg}}$. A 30 s diversion would require 63 kg and is NO-GO. Ten seconds is safe but not the requested longest safe choice.

Physical / decision gate. Choose the longest listed diversion that leaves nonnegative fuel. The terminal constant-speed model is separate from the whole-descent average; do not use that average as net terminal acceleration.

Checkpoint consequences. AMBER: the landing pad needs a last-second correction; charge 1 and land at a less desirable site. RED: fuel prediction rejects the landing; charge 2 and rehearse an immediate wave-off. If reserve is exhausted here, no landing occurs; end with a safe lunar-orbit abort.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-07 A walk with a turnaround time

Timeline / provenance: Planned EVA duration 2 h 40 min. [F1, p. 1-2]

Calculation outline — all required outputs

For each walker, oxygen starts at 1.80 kg; modeled use is 0.480 kg/h and protected reserve is 0.300 kg. Find use and remaining oxygen over the 160 min EVA. Find maximum allowed EVA duration and schedule margin. Egress and return consume 28 min of the planned EVA; each science station takes 12 min. How many complete stations fit?

Hand method and slide-rule scales. C/D: convert minutes to hours, multiply by flow, and divide usable oxygen by flow. Pencil: subtract reserves and non-science time. Round station count down, never up.

Worked answers (guard digits are not player precision)

EVA $= 160/60 = 2.66667$ h. Use $= 0.480(2.66667) =$ $\boxed{1.28 \text{ kg per walker}}$. Remaining $= 1.80 - 1.28 =$ $\boxed{0.520 \text{ kg}}$, so GO. Allowed duration $= (1.80 - 0.300)/0.480 = 3.125$ h $=$ $\boxed{187.5 \text{ min}}$ (3 h 7 min 30 s). Margin $= 187.5 - 160 =$ $\boxed{27.5 \text{ min}}$. Stations $= \lfloor (160 - 28)/12 \rfloor =$ $\boxed{11}$. Do not divide one walker's tank by the combined two-person use rate.

Physical / decision gate. Both walkers must retain at least 0.300 kg. The protected reserve is not available for science. These suit quantities and task times are invented for play.

Checkpoint consequences. AMBER: Houston calls an early turnaround; charge 1 and record "reduced science." RED: the oxygen or station plan violates a return margin; charge 2 and record "EVA curtailed." Correct the budget before ascent preparations.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-08 Rise, meet, and dock

Timeline / provenance: Planned ascent 124:23:26, 438 s; CSI 125:21:19.1. [F1, Table 1-2]

Calculation outline — all required outputs

For ascent $\Delta v = 6060$ ft/s, find m/s and average acceleration over 7 min 18 s. Find insertion GET and time until CSI. At the separate docking exercise, range is 40.0 m and closure speed 0.200 m/s. Find time to contact without braking and a burn duration to cancel closure using 0.0500 m/s^2.

Hand method and slide-rule scales. Use C/D for conversion and division, pencil for GET. For docking, $t_{\text{contact}} = r/v$ and $t_{\text{brake}} = \Delta v/a$. The nominal burn is initiated immediately; stop the modeled closure before contact and then let the referee narrate docking.

Worked answers (guard digits are not player precision)

$\Delta v = 6060(0.3048) = \boxed{1.85 \times 10^3 \text{ m/s}}$. $\bar{a} = 1847.088/438 = \boxed{4.22 \text{ m/s}^2}$. Insertion 124:23:26 + 00:07:18 = $\boxed{124:30:44}$. CSI coast 125:21:19.1 − 124:30:44 = $\boxed{00:50:35.1}$. Time to contact = 40/0.200 = $\boxed{200 \text{ s}}$. Braking = 0.200/0.0500 = $\boxed{4.00 \text{ s}}$, opposite closure. During immediate uniform braking, travel is only $\frac{1}{2}(0.200)(4) = 0.400$ m, safely below the 40 m range.

Physical / decision gate. Command braking opposite to closure. Wrong direction is RED. The close-range model is not the CSI maneuver and does not model the actual rendezvous orbit.

Checkpoint consequences. AMBER: Collins delays docking while the crew corrects its pad; charge 1. RED: the docking safety predictor flags a collision; charge 2 for an emergency separation rehearsal. Exhaustion after landing means the simulation ends with rendezvous unresolved, not an invented safe rescue.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-09 Bring Columbia home

Timeline / provenance: Planned TEI 135:24:33.8, 149.4 s; EI 195:03:27. [F1]

Calculation outline — all required outputs

For components $(3213.3, 705.0, -138.8)$ ft/s, find vector magnitude in ft/s and m/s, and average acceleration over 2 min 29.4 s. Find cutoff GET and coast time from cutoff to entry interface.

Hand method and slide-rule scales. Use A/D to square and take a root; pencil to add; C/D to convert and divide. Convert duration before adding GET. Subtract the cutoff time, not ignition, from entry interface.

Worked answers (guard digits are not player precision)

$\Sigma\Delta v_i^2 = 10{,}841{,}587.33$ $(\text{ft/s})^2$. $\Delta v = 3292.65658$ ft/s $\rightarrow$ $\boxed{3.29 \times 10^3 \text{ ft/s}}$. Multiply by 0.3048: $\boxed{1.00 \times 10^3 \text{ m/s}}$ (guard value 1003.60172). $\bar{a} = 1003.60172/149.4 = \boxed{6.72 \text{ m/s}^2}$. Cutoff = $\boxed{135{:}27{:}03.2}$. Coast 195:03:27 − 135:27:03.2 = $\boxed{59{:}36{:}23.8}$. The plan's rounded required magnitude is 3292.7 ft/s.

Physical / decision gate. Retain the signed components on the burn command. An incorrect direction is RED even with the right magnitude. Coast time is a schedule calculation, not an orbital prediction.

Checkpoint consequences. AMBER: Earth-return targeting needs a trim; charge 1. RED: the entry predictor rejects the pad; charge 2 for a return-burn replan. Exhaustion means an unresolved return simulation.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

11-10 Entry corridor and recovery

Timeline / provenance: Planned EI 195:03:27; detailed splashdown 13:55 later. [F1, p. 3-135]

Calculation outline — all required outputs

At the training gate, horizontal speed is 11.0 km/s and downward vertical speed is 1.25 km/s. Find flight-path angle $\gamma = -\arctan(v_{\text{down}}/v_{\text{horizontal}})$, speed magnitude, and kinetic energy per unit mass in MJ/kg. Check the modeled corridor $-6.8° \le \gamma \le -6.2°$ and energy limit 63.0 MJ/kg. The detailed plan gives splashdown at EI plus 13 min 55 s. Find splashdown GET and confirm the interval.

Hand method and slide-rule scales. C/D gives the speed ratio; interpolate the tangent table on the reference sheet or use T. Use A/D for speed. $e = v^2/2$: a squared speed in $(\text{km/s})^2$ gives MJ/kg directly. Use pencil for clock subtraction.

Worked answers (guard digits are not player precision)

Ratio $= 1.25/11 = 0.113636$. Table interpolation gives about 6.48°; $\boxed{\gamma = -6.48°}$. $v = \sqrt{11^2 + 1.25^2} = \boxed{11.1 \text{ km/s}}$. $e = (121 + 1.5625)/2 = \boxed{61.3 \text{ MJ/kg}}$. $\boxed{\text{GO on both gates}}$. Splashdown 195:03:27 + 00:13:55 $= \boxed{195:17:22}$. Subtracting EI confirms $\boxed{00:13:55}$. The summary rounds this to 195:17; the game uses the more precise detailed timeline rather than inventing zero seconds.

Physical / decision gate. Both corridor and energy checks must be GO. Positive γ is an upward trajectory and is RED. These limits are game thresholds, not authenticated Apollo flight rules.

Checkpoint consequences. AMBER: the predictor requests a corrected pad; charge 1. RED: a shallow angle predicts skip-out, a steep angle predicts excessive deceleration, or an energy error flags heating uncertainty; charge 2. Do not enter until corrected. With reserve intact, narrate parachutes and recovery; otherwise the return simulation is unresolved.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-01 The alarm: power the lifeboat

Timeline / provenance: Historical tank rupture 055:54:53. [N1]; this activation model is invented.

Calculation outline — all required outputs

In a separate emergency CM battery pool, usable energy is 2.40 kWh. Load is 1.20 kW and LM activation takes 22 min. Find initial endurance, activation energy, remaining energy, and endurance remaining at that load. Find activation completion GET. Is activation possible before this pool is exhausted?

Hand method and slide-rule scales. Use C/D: $t = E/P$, $E = Pt$, minutes divided by 60. Subtract energy on paper; add 22 min to the alarm time. This emergency pool is not the protected entry battery on card 13-10.

Worked answers (guard digits are not player precision)

Initial endurance $= 2.40/1.20 =$ $\boxed{2.00\text{ h}}$. Activation $= 22/60 = 0.366667$ h; energy $= 1.20(0.366667) =$ $\boxed{0.440\text{ kWh}}$. Remaining $= 2.40 - 0.440 =$ $\boxed{1.96\text{ kWh}}$. Remaining endurance $= 1.96/1.20 =$ $\boxed{1.63\text{ h}}$ (98 min). Completion 055:54:53 + 00:22:00 = $\boxed{056:16:53}$. $\boxed{\text{GO}}$: 22 min is less than 120 min.

Physical / decision gate. Activate Aquarius immediately and transfer the crew. Keeping Odyssey at this load indefinitely is NO-GO. The explosion is a scripted historical event, never a penalty for bad math.

Checkpoint consequences. AMBER: transition checklist delays threaten battery margin; charge 1. RED: the power-down budget is rejected; charge 2 for a contingency transfer rehearsal. Reserve exhaustion means the rescue simulation fails.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-02 Restore a free return

Timeline / provenance: Historical LM return maneuver near GET 061:30; magnitude 11.6 m/s. [N1]

Calculation outline — all required outputs

Training components are $(+11.0, +3.70)$ m/s. Find total Δv and burn time using average acceleration 0.400 m/s^2. For a modeled constant thrust 12,000 N, $I_{sp} = 305$ s, and $g_0 = 9.81$ m/s^2, find mass flow and propellant used.

Hand method and slide-rule scales. A/D handles vector magnitude. C/D: $t = \Delta v/\bar{a}$, $\dot{m} = F/(I_{sp}g_0)$, $m_{used} = \dot{m}t$. Keep intermediate readings on the sheet; do not repeatedly round them. The fixed average acceleration is a supplied approximation.

Worked answers (guard digits are not player precision)

$\Delta v = \sqrt{11^2 + 3.7^2} = \sqrt{134.69} = \boxed{11.6 \text{ m/s}}$. $t = 11.605602/0.400 = \boxed{29.0 \text{ s}}$. $\dot{m} = 12{,}000/(305 \cdot 9.81) = \boxed{4.01 \text{ kg/s}}$. $m_{used} = 4.010629(29.014005) = \boxed{116 \text{ kg}}$ (guard value 116.364). Remaining $= 150 - 116.364 = \boxed{33.6 \text{ kg}}$. $\boxed{\text{GO}}$, above the 20 kg reserve. Mass and thrust values are training inputs, not the historical burn pad.

Physical / decision gate. Preserve both positive components. For this isolated maneuver, usable propellant is 150 kg and at least 20 kg must remain. Check the fuel gate; it is not the PC+2 tank inventory.

Checkpoint consequences. AMBER: return tracking predicts a correction; charge 1. RED: a wrong-way or underbudgeted burn fails the free-return predictor; charge 2 for a replanning rehearsal. No lunar landing is available as an alternative ending.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-03 Three people, one oxygen budget

Timeline / provenance: Game accounting epoch 058:00:00; historical splashdown 142:54:41. [N1]

Calculation outline — all required outputs

Compute remaining flight time T in h:m:s and decimal hours. LM usable oxygen is 12.0 kg with a protected 2.00 kg reserve. Compare "conserve" use 0.0350 kg/(person h) against "normal" 0.0500 kg/(person h), for three people. Calculate total use and remaining oxygen for each; choose a safe profile.

Hand method and slide-rule scales. Subtract GET on paper. On C/D calculate $T = 84{+}54/60{+}41/3600$ and $m = 3rT$. Subtract each use from 12.0 kg. Keep T to at least four digits on paper for later cards.

Worked answers (guard digits are not player precision)

$142{:}54{:}41 - 058{:}00{:}00 = \boxed{84{:}54{:}41}$; $T = \boxed{84.9113889\ \text{h}}$, normally read as 84.9 h. Conserve: $3(0.0350)T = \boxed{8.92\ \text{kg}}$; remaining $\boxed{3.08\ \text{kg}}$ (3.084304), GO. Normal: $3(0.0500)T = \boxed{12.7\ \text{kg}}$; remaining $\boxed{-0.737\ \text{kg}}$ (minus 0.736708), NO-GO. Choose $\boxed{\text{conserve}}$. Its margin above the protected 2 kg is 1.084304 kg.

Physical / decision gate. Remaining oxygen must be at least 2.00 kg. Do not confuse per-person use with total cabin use. Carry the chosen profile to the mission ledger; all rates are fictional planning averages.

Checkpoint consequences. AMBER: an incorrect budget prompts tighter rationing; charge 1. RED: the selected profile cannot sustain the protected oxygen margin; charge 2 for an emergency conservation replan.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-04 A dark, cold electrical plan

Timeline / provenance: Same 058:00:00–142:54:41 horizon as 13-03.

Calculation outline — all required outputs

Use T from 13-03. LM energy pool is 18.0 kWh; retain 2.50 kWh. Non-idle operations cost 0.600 kWh in addition to idle use. Compare idle loads 0.160 kW (low) and 0.240 kW (normal). Find total use and remaining energy for each, and maximum permissible idle power. Choose a profile.

Hand method and slide-rule scales. C/D: $E_{\text{use}} = PT + 0.600$. Pencil: $E_{\text{left}} = 18 - E_{\text{use}}$. Then $P_{\max} = (18 - 2.50 - 0.600)/T$. Energy is kWh; power is kW. Do not multiply power by seconds here.

Worked answers (guard digits are not player precision)

Low use $= 0.160(84.9113889) + 0.600 =$ $\boxed{14.2\text{ kWh}}$ (14.185822); remaining $\boxed{3.81\text{ kWh}}$. Normal use $= 0.240T + 0.600 =$ $\boxed{21.0\text{ kWh}}$; remaining $\boxed{-2.98\text{ kWh}}$. $P_{\max} = 14.9/84.9113889 =$ $\boxed{0.175\text{ kW}}$ (175.477 W). Choose $\boxed{\text{low}}$. At 0.160 kW the reserve margin is 1.314178 kWh above 2.50.

Physical / decision gate. The low-power plan must retain at least 2.50 kWh. The 0.600 kWh operations allowance already includes modeled LM burns; do not charge them again. This pool is separate from CM entry energy.

Checkpoint consequences. AMBER: incorrect switching instructions force load shedding; charge 1. RED: the chosen profile predicts depletion before recovery; charge 2 and rewrite the electrical plan. Cold and dim conditions are narrative effects, not additional hidden resource deductions.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-05 PC+2: shorten the trip

Timeline / provenance: Historical PC+2 ignition 079:27:39; burn 263.4 s, $\Delta v = 262$ m/s. [N1]

Calculation outline — all required outputs

For a fictional mass model, $m_0 = 30{,}000$ kg, effective exhaust velocity $v_e = 3000$ m/s, target $\Delta v = 262$ m/s, and average propellant flow 9.525 kg/s. Find final mass, propellant used, and burn duration. Find historical scheduled cutoff using 263.4 s, not your rounded prediction.

Hand method and slide-rule scales. Use C/D and L: $x = \Delta v/(2.302585 v_e)$, $m_f = m_0\, 10^{-x}$, $m_{\text{used}} = m_0 - m_f$, $t = m_{\text{used}}/\dot{m}$. On L, find the mantissa $1 - x$, read 10^{1-x} on D, then divide by 10 to obtain 10^{-x}. Subtraction uses pencil; retain extra readable digits before subtracting.

Worked answers (guard digits are not player precision)

$x = 262/(2.302585 \cdot 3000) = 0.03792838$. $10^{-x} = 0.91637159$. $m_f = 30{,}000(0.91637159) = \boxed{27{,}491.15 \text{ kg}}$ (slide-rule reading about 27,490). Used $= 30{,}000 - 27{,}491.15 = \boxed{2508.85 \text{ kg}}$, report 2.51×10^3 kg. $t = 2508.85237/9.525 = \boxed{263 \text{ s}}$ (263.39657 s). Remaining burn pool $= 2700 - 2508.85 = \boxed{191 \text{ kg}}$, GO. Historical cutoff 079:27:39 + 00:04:23.4 = $\boxed{079{:}32{:}02.4}$. Allow the normal 1% band on propellant despite subtractive sensitivity.

Physical / decision gate. The isolated burn's usable propellant is 2700 kg; retain 150 kg. Mass must decrease, so $m_f < m_0$. A wrong exponent sign is RED. The historical time is an anchor; the masses and flow are invented.

Checkpoint consequences. AMBER: duration disagrees with the acceleration plan; charge 1 for a throttle-pad correction. RED: the burn budget or exponent is wrong; charge 2 for an emergency targeting rehearsal.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-06 Cooling water versus drinking water

Timeline / provenance: Same full return horizon $T = 84.9113889$ h; planning epoch 058:00:00.

Calculation outline — all required outputs

Water starts at 100 kg; protect 12.0 kg. Each of three crew uses 0.0600 kg/h for drinking. Compare cooling 0.900 kg/h (normal) with 0.700 kg/h (reduced). Find combined rates, total use, and water left for both profiles. Find the maximum allowable cooling rate, then choose a safe profile.

Hand method and slide-rule scales. C/D: $r_{\text{total}} = r_{\text{cool}} + 3r_{\text{drink}}$, $m_{\text{use}} = r_{\text{total}}T$. Subtract from 100 on paper. $r_{\text{cool,max}} = (100 - 12)/T - 3r_{\text{drink}}$.

Worked answers (guard digits are not player precision)

Drinking rate $= 3(0.0600) = 0.180$ kg/h. Normal total $=$ $\boxed{1.08 \text{ kg/h}}$; use $= 1.08T =$ $\boxed{91.7 \text{ kg}}$; left $\boxed{8.30 \text{ kg}}$, NO-GO. Reduced total $=$ $\boxed{0.880 \text{ kg/h}}$; use $= 0.880T =$ $\boxed{74.7 \text{ kg}}$; left $\boxed{25.3 \text{ kg}}$, GO. Maximum cooling $= 88/T - 0.180 =$ $\boxed{0.856 \text{ kg/h}}$. Choose $\boxed{\text{reduced}}$. Its margin above reserve is 13.277978 kg.

Physical / decision gate. Keep at least 12.0 kg. Both proposed cooling settings are assumed thermally permissible in this game; no thermal engineering calculation is implied. The drinking allowance is not optional.

Checkpoint consequences. AMBER: a bad ration sheet prompts an uncomfortable reduced-cooling replan; charge 1. RED: the selected profile consumes protected water; charge 2 for emergency ration planning. Do not invent additional water losses for narrative delays.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-07 Make the square cartridge fit

Timeline / provenance: Training checkpoint GET 090:00:00; CO_2 planning to 134:00:00.

Calculation outline — all required outputs

Three people produce 0.900 kg of CO_2 per person per day. Existing compatible absorbent has 0.600 kg capacity remaining. Each adapted spare removes 1.80 kg. Find hourly production, the load for the next 44 h, and the minimum integer number of spares needed in addition to existing capacity. Find capacity margin, time until existing capacity runs out, and the adapter deadline GET. Assembly takes 2.00 h: find time slack.

Hand method and slide-rule scales. Use C/D for rate conversion and load. Compute $n = \lceil (m_{\text{load}} - 0.600)/1.80 \rceil$: round UP. Deadline $t = 0.600/r_{\text{hourly}}$. Add deadline to 090:00:00 and subtract assembly time on paper.

Worked answers (guard digits are not player precision)

Production $= 3(0.900)/24 = \boxed{0.1125 \text{ kg/h}}$. Load $= 0.1125(44) = \boxed{4.95 \text{ kg}}$. $n = \lceil (4.95 - 0.600)/1.80 \rceil = \lceil 2.41667 \rceil = \boxed{3 \text{ spares}}$. Capacity margin $= 0.600 + 3(1.80) - 4.95 = \boxed{1.05 \text{ kg}}$. Existing endurance $= 0.600/0.1125 = \boxed{5.33333 \text{ h}}$ (5 h 20 min). Deadline $= \boxed{095{:}20{:}00}$. Assembly slack $= 5.33333 - 2 = \boxed{3.33333 \text{ h}}$ (3 h 20 min). GO; finish by 092:00:00 if starting now.

Physical / decision gate. Finish the adapter before existing capacity expires and allocate enough spares. Timing this card uses decimal endurance bands and clock bands; cartridge count must be the exact minimum integer.

Checkpoint consequences. AMBER: a miscount requires inventory review; charge 1. RED: the predicted scrubber gap would allow CO_2 accumulation; charge 2 for a rushed adapter rehearsal. The referee must not allow the game to proceed with a knowingly inadequate scrubber plan.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-08 Navigate without the usual computer

Timeline / provenance: Historical correction near GET 105:19 used about a 15 s burn. [N1]

Calculation outline — all required outputs

A fictional two-axis correction requires $(\Delta v_x, \Delta v_y) = (-2.30, +0.400)$ m/s. Find magnitude and burn time at average acceleration 0.155 m/s^2. Find angle above the negative x direction: $\theta = \arctan(0.400/2.30)$. Write the commanded quadrant explicitly.

Hand method and slide-rule scales. Use A/D for the vector root and C/D for division. Use T, or interpolate the tangent table supplied on the reference sheet. Signs define the quadrant; θ is a small positive angle measured from the negative x axis toward positive y, not from positive x.

Worked answers (guard digits are not player precision)

$\Delta v = \sqrt{2.30^2 + 0.400^2} = \sqrt{5.45} = \boxed{2.33 \text{ m/s}}$. $t = 2.3345235/0.155 = \boxed{15.1 \text{ s}}$. Ratio $= 0.400/2.30 = 0.173913$. T gives $\boxed{9.87^\circ}$. Table interpolation: $9.5 + (0.173913 - 0.16734)/(0.17633 - 0.16734)(0.5) = 9.866^\circ$. Command $\boxed{\text{negative } x,\ \text{positive } y}$, or about 170.13° counterclockwise from positive x. The historical maneuver's components are not being reconstructed.

Physical / decision gate. Angle bands are GREEN within 0.05°, AMBER through 0.20°, RED beyond. Wrong quadrant is always RED. Magnitude and time still use ordinary relative-error bands.

Checkpoint consequences. AMBER: the predictor flags aim drift; charge 1 for an attitude reset. RED: a wrong quadrant or angle predicts a missed entry corridor; charge 2 for a complete manual-navigation rehearsal.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-09 Thread the entry corridor

Timeline / provenance: Final historical correction lasted 22.4 s; game entry predictor is invented. [N1]

Calculation outline — all required outputs

Current modeled entry angle is $-5.90°$; target is $-6.50°$. A positive corrective Δv changes angle by $-0.0400°$ per m/s. Burn acceleration is 0.670 m/s^2. Find required Δv, nominal burn duration, and predicted angle. Find the minimum and maximum burn durations allowed by $-6.80° \le \gamma \le -6.20°$.

Hand method and slide-rule scales. Use signed subtraction for $\Delta\gamma$. C/D: $\Delta v = (\gamma_{\text{target}} - \gamma_0)/k$ and $t = \Delta v/a$. Repeat for both corridor edges. Use $\gamma(t) = \gamma_0 + kat$ to check signs.

Worked answers (guard digits are not player precision)

$\Delta\gamma = -6.50 - (-5.90) = -0.600°$. $\Delta v = (-0.600)/(-0.0400) = \boxed{15.0 \text{ m/s}}$. $t = 15/0.670 = \boxed{22.4 \text{ s}}$; $\gamma = -5.90 - 0.0400(15) = \boxed{-6.50°}$. Shallow edge: $\Delta v = 7.50$ m/s, $t_{\min} = \boxed{11.2 \text{ s}}$ (11.19403). Steep edge: $\Delta v = 22.5$ m/s, $t_{\max} = \boxed{33.6 \text{ s}}$ (33.58209). Thus $11.19403 \le t \le 33.58209$ s is the toy corridor. The 22.4 s historical agreement is a chosen training calibration.

Physical / decision gate. Command a positive correction with a negative angle response. The nominal pad must target $-6.50°$, not merely any duration in the corridor. Angle bands are $0.05°/0.20°$. Report the two duration bounds in ascending order.

Checkpoint consequences. AMBER: a small duration error produces a corridor warning; charge 1. RED: a too-short burn predicts shallow skip-out; too long predicts excessive entry loading. Charge 2 and correct before separation.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

13-10 Wake Odyssey and come home

Timeline / provenance: Historical LM jettison 141:30:02; splashdown 142:54:41. [N1]

Calculation outline — all required outputs

A separate, protected CM battery starts with 3.00 kWh. Reactivation uses 0.800 kW for 45 min; entry 1.20 kW for 30 min; recovery beacon 0.150 kW for 4.00 h. Find each energy, total use, and energy remaining; retain 0.500 kWh. Reactivation begins at 140:30:02: find completion GET and margin until LM jettison. Find LM-jettison-to-splashdown interval.

Hand method and slide-rule scales. Convert minutes to hours on C/D, then use $E = Pt$. Add energy with pencil; subtract from 3.00. Add 45 min to reactivation GET and subtract the historical event times. Beacon use extends after splashdown and is included in this pool.

Worked answers (guard digits are not player precision)

Reactivation $= 0.800(45/60) =$ $\boxed{0.600\ \text{kWh}}$. Entry $= 1.20(30/60) =$ $\boxed{0.600\ \text{kWh}}$. Beacon $= 0.150(4) =$ $\boxed{0.600\ \text{kWh}}$. Total $=$ $\boxed{1.80\ \text{kWh}}$; left $= 3.00 - 1.80 =$ $\boxed{1.20\ \text{kWh}}$, GO. Completion 140:30:02 + 00:45:00 $=$ $\boxed{141:15:02}$. Jettison margin $=$ $\boxed{00:15:00}$, GO. Jettison to splashdown 142:54:41 − 141:30:02 $=$ $\boxed{01:24:39}$. The power schedule is fictional; the separation and recovery anchors come from the NASA catalog.

Physical / decision gate. Energy remaining must be at least 0.500 kWh and reactivation must finish before 141:30:02. The CM battery was not charged for the LM's earlier loads; do not merge those inventories.

Checkpoint consequences. AMBER: a sequencing error threatens the transfer; charge 1. RED: premature jettison or insufficient entry energy fails the recovery predictor; charge 2 and correct the sequence. If reserve remains nonnegative and all gates are corrected, narrate parachutes and recovery of all three crew. Otherwise end with rescue unproven.

Grade the first committed sheet using the referee protocol. Ordinary readings use 1% / 5% bands; time arithmetic and angles use their own bands. Use the worst result, including the decision gate. Charge reserve once, then require a corrected sheet before continuing.

Mission Control narration: how to use these scripts

Original game dialogue — not historical quotations

Everything in quotation marks in this appendix is invented read-aloud dialogue or scene description. It is not a NASA transcript. Use it verbatim or adapt it to your crew, without changing the inputs, grading bands, resource rules, or historical event anchors.

Before a card. Read its briefing below, announce the GET printed on the calculation card, and read that card's telemetry and orders. Where the card supplies no exact event time, name the phase instead; do not invent a new deadline. Briefings intentionally omit answers.

After commitment. Grade first, subtract the appropriate command-reserve charge once, and check the new reserve. Read only the matching GREEN, AMBER, or RED example. For an error, the examples below assume reserve is still nonnegative. If reserve is negative, use the appropriate ending instead of inviting another attempt. A correct sheet never receives a scripted error.

Correction and continuation. GREEN narration authorizes the nominal game action. AMBER and RED narration hold that action pending a corrected pad. After correction, use the common release line below, then the card's GREEN scene where appropriate. For 11-06 AMBER, retain the less desirable landing site; for 11-07 retain the recorded science flag rather than announcing full science.

Reusable referee lines

Commitment: "Flight crew, give me your signed command, units, and GO or NO-GO decision. Is that your committed pad?"
AMBER bookkeeping: "That is an AMBER checkpoint. One command-reserve point is charged. Your reserve is now [read the ledger]. Revise the pad before we proceed."
RED bookkeeping: "That is a RED checkpoint. Two command-reserve points are charged, once for this sheet. Your reserve is now [read the ledger]."
Release after correction: "The corrected pad is accepted. There is no additional charge for this revision. Resume from this card's nominal calculation baseline; keep the warning on the ledger."
Unmodeled passage of time: "We now advance to the next card's supplied epoch. The intervening routine work is narrated; no extra consumables are deducted."

Predictions are not actual disasters. Error scenes describe the simulator's warning or contingency rehearsal. Do not narrate a collision, exhausted tank, or atmospheric entry as having occurred while players are still allowed to correct their commands. No scene adds dice rolls, real-time pressure, hidden fuel charges, or extra calculations. The cartridge scene is dramatic description, not a breathing-apparatus construction procedure.

Apollo 11: launch and narrated transitions

Opening / before 11-01. “Armstrong, Aldrin, Collins: your mission is to land Eagle, complete the planned surface work, and bring the crew home. The Saturn rises through the cloud layer. Staging is nominal; Earth curves beneath you. At planned GET 000:11:43, the spacecraft reaches its parking orbit. For this game the launch is successful. Houston is preparing your lunar-departure pad. Take out your pencils and slide rule.”

After corrected 11-01 / before 11-02: docking and extraction. “The third-stage engine shuts down and the cabin settles into weightlessness. Collins turns Columbia to face Eagle, eases toward the docking target, and secures the connection. The lunar module is extracted from the stage. That routine docking is narrated here, not a new calculation checkpoint. Now the spent stage must be left safely behind: prepare the avoidance pad.”

After corrected 11-02 / before 11-03: translunar coast. “The spent stage recedes. Earth becomes a smaller blue disk while the Moon grows ahead. The crew alternates navigation observations, housekeeping, meals, and rest. Routine midcourse updates and thermal control are handled in the narrated coast. Houston now sends the lunar-capture pad. Check it before the Moon cuts off our radio link.”

After corrected 11-03 / before 11-04: orbit operations. “Radio contact returns as Columbia comes around the Moon. For the game’s accepted capture branch, you are in lunar orbit. The second orbit-insertion maneuver and routine LM inspection are completed as narrated flight-plan operations. Collins has daylight observations and communications to schedule; these are not the same kind of window.”

After corrected 11-04 / before 11-05: undock. “Armstrong and Aldrin complete Eagle’s checkout. Collins remains in Columbia, monitoring the separation. The docking connection releases and the two vehicles drift apart. Separation and activation are narrated operations here. Eagle’s next checkpoint is the burn that lowers its perilune.”

After corrected 11-05 / before 11-06: descent readiness. “Eagle coasts toward the powered-descent point. Aldrin reviews the corrected timing while Armstrong watches the surface move below. Houston calls for the final descent-readiness check. No new coast budget is charged. At the next card’s terminal exercise, decide how much diversion the supplied fuel model permits.”

After corrected 11-06 / before 11-07: surface operations. “Eagle is settled on the surface in the accepted landing branch. The crew completes the contact checks, secures the spacecraft, and prepares for the planned walk. Collins continues his orbiting watch. Before either walker leaves, Houston wants the oxygen and science schedule checked.”

After corrected 11-07 / before 11-08: prepare ascent. “The walkers return and secure the cabin. Record the science result already earned; do not restore curtailed work. Equipment stowage, rest, and ascent preparations are narrated. Collins is ready in Columbia. Eagle’s crew must now check ascent timing and the separate close-range braking exercise.”

After corrected 11-08 / before 11-09: jettison. “The corrected approach brings the docking targets together. Contact, capture, and a secure connection are confirmed. Armstrong and Aldrin transfer back to Columbia. The LM-jettison sequence and separation maneuver are narrated without another calculation. All three crew now prepare the trans-Earth injection pad.”

Apollo 13: launch, crisis, and narrated transitions

Opening / before 13-01. "Lovell, Haise, Swigert: the original mission is a lunar landing. Launch, Earth-orbit checkout, lunar departure, and docking are successful for this game. Odyssey and Aquarius head outward together. Routine coast operations pass without a calculation checkpoint. Now announce GET 055:54:53: a sharp disturbance runs through the spacecraft, and warning indications appear on the panels. Something has failed in the service module. The landing mission is over; your new mission is to bring three people home."

After corrected 13-01 / before 13-02: transfer. "Odyssey's nonessential loads go dark. Haise brings Aquarius into its lifeboat role while Swigert preserves the command module for later. The crew moves through the connecting tunnel with checklists and essential equipment. Houston now needs a return maneuver that the LM engine can perform with the two spacecraft still docked."

After corrected 13-02 / before 13-03–04: planning audit. "The accepted correction puts the simulator on its return branch. Houston and the crew spread out the consumable sheets. The oxygen and power audits both start their accounting at 058:00:00, even though this discussion follows the maneuver scene. They cover the whole remaining trip; they are not fresh snapshots or deductions from an already-used inventory."

After corrected 13-04 / before 13-05: lunar swingby. "Aquarius is quiet and dim under the accepted conservation plan. The Moon fills the windows and then passes behind the docked stack. For a time there is no radio contact. When Houston's voice returns, the crew prepares PC+2: the major burn intended to shorten the trip and improve the return target. Read the mass model carefully."

After corrected 13-05 / before 13-06: return coast. "The burn ends and Earth becomes the destination ahead. The crew settles back into the low-power return coast. Houston now reviews cooling and drinking water. This is another audit of the same full horizon starting at 058:00:00, not an extra charge for the time just narrated."

At 13-07: cartridge-adapter assembly. Read the briefing and telemetry, then grade the committed plan. Only after the plan is accepted or corrected, read: "Haise lays out the spare cartridges and the materials available in the cabin. Swigert reads the simulated ground instructions while Lovell holds the parts steady in weightlessness. The crew assembles the adapter during the card's stated assembly interval. For this game, the adapter connects the allocated cartridges to the airflow path and the capacity plan is accepted. No extra crafting roll or chemical calculation is required."

After corrected 13-07 / before 13-08–09: manual navigation. "The adapted scrubber arrangement is working in the accepted game branch. With the ordinary guidance setup unavailable for this exercise, Lovell uses the visible Earth as an attitude reference. Haise reads the signed correction pad, and Swigert checks the units. The later entry predictor will ask a different question: how long must the final correction last?"

After corrected 13-09 / before 13-10: ready Odyssey. "The corrected entry pad is accepted. The damaged service module is separated in the narrated sequence. Swigert prepares to wake Odyssey while Aquarius is still available. Do not jettison the lifeboat until the final card's energy and reactivation timing gates have been accepted."

Apollo 11 checkpoint scripts / 11-01–02

11-01 — Commit to the Moon

Briefing (before calculation): "Houston to Columbia: we are at the lunar-departure gate. Aldrin, read the velocity units. Collins, check the duration and decimal place. Armstrong, hold the command until the crew commits."

GREEN (accepted command): "The departure pad is accepted. The third stage lights, acceleration builds, and the burn ends on the checked clock. Columbia and Eagle begin the game's outbound lunar coast."

AMBER (charge 1, then correct): "Guidance review has flagged a targeting discrepancy. We need a corrective update before accepting the departure pad. Check your unit conversion, average acceleration, and cutoff time."

RED (charge 2; if reserve permits, then correct): "The simulator rejects this departure solution. Run the abort-and-replan rehearsal; do not execute this pad. If command reserve permits, rebuild the calculation and read it back."

11-02 — Clear the spent stage

Briefing (before calculation): "Columbia, the docked spacecraft is clear of the extraction sequence, but the stage is still nearby. Read the vector as one maneuver, then tell Houston its magnitude and commanded duration."

GREEN (accepted command): "The safe-separation predictor accepts your pad. The short maneuver sends the spent stage steadily away. The crew can turn its attention to the coast ahead."

AMBER (charge 1, then correct): "The predicted separation rate differs from the accepted model. Houston requests a tracking update and corrected magnitude. Keep the component directions on the revised pad."

RED (charge 2; if reserve permits, then correct): "The safe-separation check fails. Rehearse emergency avoidance in the simulator; no collision has occurred in the game. Rebuild the vector and duration before proceeding."

Apollo 11 checkpoint scripts / 11-03–04

11-03 — Capture lunar orbit

Briefing (before calculation): "The Moon is about to block Houston's signal. Collins, verify the retrograde signs; Aldrin, check the magnitude and cutoff clock. Your committed pad must stand on its own."

GREEN (accepted command): "The engine shuts down at the accepted cutoff. As the spacecraft emerges from behind the Moon, radio contact returns. The simulator accepts lunar capture."

AMBER (charge 1, then correct): "Tracking predicts an off-target lunar orbit. Houston requests a trim update. Correct the signed pad and timing before we accept the nominal capture branch."

RED (charge 2; if reserve permits, then correct): "The simulator rejects lunar capture with this pad. Prepare the contingency-return rehearsal. If reserve remains, correct the burn solution before resuming."

11-04 — Work the lunar clock

Briefing (before calculation): "Collins, sunlight and radio coverage are separate constraints. Houston needs the observation schedule checked against daylight, not against a coverage duration that happens to look similar."

GREEN (accepted command): "The daylight task fits, and the schedule is accepted. Record the coverage fraction separately; we have not assumed that the entire observation has continuous radio contact."

AMBER (charge 1, then correct): "The observation timing would lose useful coverage. Houston requests a revised schedule. Recheck the orbit period, daylight interval, and coverage percentage independently."

RED (charge 2; if reserve permits, then correct): "The handover schedule is rejected. Rebuild the scheduling sheet; do not substitute a radio window for a daylight window. The missed handover is a simulator contingency."

Apollo 11 checkpoint scripts / 11-05–06

11-05 — Lower Eagle's perilune

Briefing (before calculation): "Eagle, Houston wants descent-orbit insertion checked before ignition. Use the required velocity change specified on this card, then tell us when cutoff occurs and how much coast remains before PDI."

GREEN (accepted command): "The DOI pad is accepted. The burn ends on the corrected clock, and Eagle enters the game's descent coast with its readiness interval accounted for."

AMBER (charge 1, then correct): "The coast briefing would be late on this sheet. Expedite the pad review in the narrative, then correct the cutoff and coast interval. No extra fuel or elapsed-time charge is added."

RED (charge 2; if reserve permits, then correct): "The descent-readiness check fails. Rehearse the full readiness sequence before attempting PDI. If reserve remains, correct the sheet and resume the nominal coast."

11-06 — Land or wave off

Briefing (before calculation): "Armstrong sees a rough patch ahead; Aldrin watches altitude and fuel. Check the supplied terminal model and choose the longest listed diversion that still permits descent without exhausting the fuel."

GREEN (accepted command): "The chosen diversion passes the fuel gate. Eagle moves toward the selected patch, descends, and settles. The crew reports contact and secures the spacecraft."

AMBER (charge 1, then correct): "The landing pad needs a last-second correction. Correct the calculation before touchdown is accepted. In the accepted continuation, Eagle lands at a less desirable site; the warning remains on the ledger."

RED (charge 2; if reserve permits, then correct): "The fuel predictor rejects the landing command. Rehearse an immediate wave-off; the game has not committed touchdown. If reserve remains, correct the diversion choice and landing pad."

Apollo 11 checkpoint scripts / 11-07–08

11-07 — A walk with a turnaround time

Briefing (before calculation): "The hatch is ready, but Houston wants the return margin protected. Budget each walker's oxygen separately, then check how many complete science stations fit the planned EVA."

GREEN (accepted command): "The EVA plan is accepted. The walkers work through the planned science schedule and return with the protected margin intact. Record full planned science if no warning occurred on this card."

AMBER (charge 1, then correct): "Houston calls for an early turnaround. Record reduced science, then correct the oxygen and station sheet. The crew returns safely in the accepted continuation, but the lost science is not restored by the correction."

RED (charge 2; if reserve permits, then correct): "The plan violates the modeled return margin. Record EVA curtailed and rehearse the immediate return. Correct the sheet before ascent preparations; do not claim that the original full science schedule was completed."

11-08 — Rise, meet, and dock

Briefing (before calculation): "Eagle's ascent pad is on the desk. Check insertion timing first. Then work the separate close-range exercise: how long before contact, and which way must you brake?"

GREEN (accepted command): "Ascent timing and braking are accepted. Eagle reaches the game's rendezvous branch; the closing motion is canceled. The docking targets align, contact is made, and the connection is secured."

AMBER (charge 1, then correct): "Collins holds the docking approach while the crew corrects its pad. Check the insertion clock and braking duration. The narrative delay adds no hidden propellant charge."

RED (charge 2; if reserve permits, then correct): "The docking predictor flags a possible collision. Rehearse emergency separation in the simulator; no impact is declared. Correct the braking direction and pad before docking."

Apollo 11 checkpoint scripts / 11-09–10

11-09 — Bring Columbia home

Briefing (before calculation): “All three crew are back aboard Columbia. Houston has supplied the Earth-return components and entry clock. Read the signs, compute the magnitude, and measure coast from cutoff.”

GREEN (accepted command): “The return pad is accepted. The engine fires and shuts down on the checked clock. The simulator advances Columbia onto its nominal Earth-return branch.”

AMBER (charge 1, then correct): “Return targeting calls for a trim update. Review the vector magnitude, acceleration, and cutoff-to-entry interval before we accept the corrected departure.”

RED (charge 2; if reserve permits, then correct): “The entry predictor rejects the return pad. Rehearse a return-burn replan and correct the signed command. Do not announce a successful Earth return until the later gates pass.”

11-10 — Entry corridor and recovery

Briefing (before calculation): “Earth fills the window. Houston’s training predictor supplies horizontal and downward speeds. Check both the angle corridor and energy gate before the game commits entry.”

GREEN (accepted command): “Both entry checks pass. In the accepted branch, the capsule descends, the parachutes open, and Columbia reaches the water on the planned game schedule. Read the successful Apollo 11 ending once all corrected gates pass.”

AMBER (charge 1, then correct): “The predictor requests a corrected entry pad. Hold the entry narration while you revise the angle, energy, or clock result. Recovery is narrated only after acceptance.”

RED (charge 2; if reserve permits, then correct): “This pad is not accepted for entry. A shallow-angle error warns of skip-out; a steep-angle error warns of excessive deceleration; an energy error raises heating uncertainty. Use only the warning that fits this sheet, then correct if reserve permits.”

Apollo 13 checkpoint scripts / 13-01–02

13-01 — The alarm: power the lifeboat

Briefing (before calculation): "Warning lights are appearing after the service-module accident. Swigert, check the emergency battery endurance. Haise, prepare Aquarius. Lovell, authorize the transfer only after the activation and power-down plan has been checked."

GREEN (accepted command): "The transfer plan is accepted. Odyssey's nonessential loads are shut down, Aquarius comes online, and the crew moves into the lifeboat on the checked schedule."

AMBER (charge 1, then correct): "The transition checklist threatens the battery margin. Revise the activation and remaining-energy figures before the transfer plan is accepted. The accident itself is not your penalty."

RED (charge 2; if reserve permits, then correct): "The power-down budget is rejected. Rehearse the contingency transfer and correct the pad. Keeping Odyssey at the emergency load indefinitely is not a rescue plan."

13-02 — Restore a free return

Briefing (before calculation): "Houston needs a return maneuver with the spacecraft still docked. Haise, work the two-axis pad; Lovell, check direction. Swigert, confirm that the modeled fuel reserve survives the burn."

GREEN (accepted command): "The free-return pad passes the model's checks. Aquarius performs the accepted correction. Houston can now plan the remaining rescue around this return branch."

AMBER (charge 1, then correct): "Return tracking predicts a correction. Review the vector magnitude, burn duration, and fuel estimate before accepting the nominal maneuver."

RED (charge 2; if reserve permits, then correct): "The free-return predictor rejects this command. Rehearse the targeting replan, preserving the supplied directions. There is no lunar-landing alternative in this rescue mission."

Apollo 13 checkpoint scripts / 13-03–04

13-03 — Three people, one oxygen budget

Briefing (before calculation): "Aquarius must support three people for the remaining trip. These are full-horizon planning figures from the stated epoch. Compare both profiles and choose one that protects the oxygen reserve."

GREEN (accepted command): "The conservation profile and budget are accepted. The ledger shows oxygen protected for the return. Keep this corrected plan; do not subtract the full horizon a second time."

AMBER (charge 1, then correct): "Houston asks for a tighter ration-sheet review. Correct the time horizon and total crew use before proceeding. The warning does not create a new oxygen loss."

RED (charge 2; if reserve permits, then correct): "The selected profile cannot protect the modeled oxygen margin. Replan conservation immediately in the simulator. A profile that depletes the pool is not accepted just because its arithmetic is written neatly."

13-04 — A dark, cold electrical plan

Briefing (before calculation): "The cabin can be quiet and dim, but the accepted plan must keep energy protected. Separate idle power from the operations allowance and check the whole return horizon."

GREEN (accepted command): "The low-power plan is accepted. Unneeded loads stay off while the required systems remain in the game plan. The crew settles into the dim return coast."

AMBER (charge 1, then correct): "The switching sheet needs load shedding. Correct the power and energy budget; keep the reserve protected. The colder, darker cabin is narration, not a second numerical penalty."

RED (charge 2; if reserve permits, then correct): "The chosen profile predicts depletion before recovery. Rewrite the electrical plan before continuing. Do not borrow the separate command-module entry battery to hide the error."

Apollo 13 checkpoint scripts / 13-05–06

13-05 — PC+2: shorten the trip

Briefing (before calculation): "The lunar swingby is behind you. Houston needs PC+2 checked with the supplied mass model. Use the logarithmic scale carefully: propellant use must reduce mass."

GREEN (accepted command): "The burn budget is accepted. Aquarius fires through the planned PC+2 interval and shuts down at the historical schedule anchor. The game's return branch now continues toward Earth."

AMBER (charge 1, then correct): "The duration differs from the accepted model. Houston requests a throttle-pad correction. Recheck the logarithm, subtraction, and division before the burn is accepted."

RED (charge 2; if reserve permits, then correct): "The burn budget or exponent fails the targeting check. Rehearse emergency targeting and rebuild the mass calculation. Do not command a model in which the spacecraft gains mass by burning fuel."

13-06 — Cooling water versus drinking water

Briefing (before calculation): "Haise has cooling requirements on one sheet and drinking allowance on another. Combine the rates for the entire return horizon without spending the water protected for entry and recovery."

GREEN (accepted command): "The reduced-cooling profile passes the game water gate. Record the accepted total and remaining water. The crew continues with the card's drinking allowance preserved."

AMBER (charge 1, then correct): "The ration sheet requires an uncomfortable reduced-cooling replan. Correct the rates and remaining water before proceeding. The narrative does not subtract additional water."

RED (charge 2; if reserve permits, then correct): "The selected profile consumes protected water. Rehearse emergency ration planning, then correct the cooling choice. The three-person drinking allowance cannot be erased to make the sheet pass."

Apollo 13 checkpoint scripts / 13-07–08

13-07 — Make the square cartridge fit

Briefing (before calculation): "Houston has an adapter concept for the available cartridges. Before we narrate assembly, count the required spares and determine when the existing capacity runs out. Round capacity needs upward."

GREEN (accepted command): "The cartridge allocation and assembly timing pass. Read the adapter-assembly transition scene. The accepted capacity plan covers this card's stated interval."

AMBER (charge 1, then correct): "The inventory sheet needs review. Recheck the production rate, allocation, or deadline before assembly is accepted. A wrong minimum integer count is RED, not merely AMBER."

RED (charge 2; if reserve permits, then correct): "The plan fails the allocation or timing gate. A scrubber gap would allow carbon-dioxide accumulation in the model. Rehearse the rushed adapter contingency, then correct the plan before narrating successful assembly."

13-08 — Navigate without the usual computer

Briefing (before calculation): "Lovell has Earth in view as the exercise's attitude reference. Haise reads the correction pad. The angle is measured from negative x toward positive y; check the quadrant before committing."

GREEN (accepted command): "The signed manual-navigation pad is accepted. Lovell holds the reference while Haise calls the burn interval. The simulator accepts the correction toward the entry target."

AMBER (charge 1, then correct): "The aim predictor flags drift. Reset the modeled attitude and correct the angle or burn duration. Do not change the reference axis to make the result look smaller."

RED (charge 2; if reserve permits, then correct): "The predictor rejects the direction or angle. A wrong quadrant could miss the entry corridor. Rehearse manual navigation from the reference again and correct the signed pad before continuing."

Apollo 13 checkpoint scripts / 13-09–10

13-09 — Thread the entry corridor

Briefing (before calculation): "Houston's local predictor says the current approach needs adjustment. Find the nominal correction and the two corridor bounds. The supplied angle response is negative for a positive correction."

GREEN (accepted command): "The nominal pad targets the accepted entry angle. The simulator accepts the final correction. The crew can now prepare Odyssey's reactivation sequence."

AMBER (charge 1, then correct): "The duration produces a corridor warning. Correct the nominal pad and both bounds before proceeding. Being somewhere inside the toy corridor does not replace the requested target."

RED (charge 2; if reserve permits, then correct): "The correction pad is rejected. Too little modeled correction warns of a shallow approach; too much warns of excessive entry loading. Select the applicable warning and rework the signed duration solution."

13-10 — Wake Odyssey and come home

Briefing (before calculation): "Swigert is ready to power Odyssey back up. Haise and Lovell check the protected battery and completion clock. Aquarius must remain attached until the accepted reactivation is complete."

GREEN (accepted command): "The energy and timing gates pass. Odyssey is ready before Aquarius is released. In the accepted branch, the capsule enters, parachutes deploy, and the crew reaches the water. Read the successful rescue ending."

AMBER (charge 1, then correct): "The transfer sequence needs correction. Hold the jettison narration while you revise the energy or timing sheet. No successful recovery is announced until the gates are accepted."

RED (charge 2; if reserve permits, then correct): "The recovery predictor rejects the sequence or battery plan. Premature jettison or insufficient entry energy would compromise the modeled return. Correct the plan if reserve remains; otherwise use the failed-rescue simulation ending."

Mission endings: read only the earned branch

Apollo 11 / all ten corrected gates; $R \geq 0$. "Columbia is on the water. Recovery crews reach the capsule and the flight crew is brought aboard. Eagle landed, and the crew came home. Your mission result is crew recovered; landing achieved." Then read exactly one science result:

- No 11-07 warning: "The record shows full planned science."
- 11-07 AMBER: "An early turnaround reduced the science return; the record keeps that warning despite the corrected budget."
- 11-07 RED: "The EVA was curtailed to protect the return margin; the record does not claim the full science schedule."

Apollo 11 / $R < 0$ on 11-01 or 11-02. "Command reserve is exhausted at the departure or avoidance checkpoint. Stop the scored run. The training scenario takes its safe Earth-orbit abort ending; the lunar landing is canceled. This is a narrated game branch, not a calculated return trajectory."

Apollo 11 / $R < 0$ on 11-03 through 11-06. "Command reserve is exhausted before an accepted landing. Houston cancels the descent mission. The training scenario takes its safe-return contingency ending; Eagle does not land. No later calculation can convert this scored abort into a completed mission." For 11-05 or 11-06, optionally add: "The scenario narrates Eagle returning to Columbia rather than continuing the landing attempt."

Apollo 11 / $R < 0$ on 11-07 through 11-10. "The simulation stops here with command reserve exhausted. We have not established a completed rendezvous or return, as applicable to this checkpoint. Record simulation terminated; return or rendezvous unresolved. Do not narrate an automatic rescue of a landed crew."

Apollo 13 / all ten corrected gates; $R \geq 0$. "Odyssey is on the Pacific. Recovery personnel reach the capsule, and Lovell, Haise, and Swigert are brought to safety. There was no lunar landing, but the rescue mission achieved its goal: all three crew recovered. Record the remaining command reserve as your team's command-quality result, not a survival probability."

Apollo 13 / $R < 0$ at any checkpoint. "Command reserve is exhausted. Stop the rescue simulation at this checkpoint; the game has not established crew survival. Record rescue simulation failed; crew survival not established. This result does not change history: the actual Apollo 13 crew returned safely."

Sources, provenance, and modeling boundaries

[F1] Supplied Apollo 11 flight-plan resources. NASA Manned Spacecraft Center, *Apollo 11, AS-506/CSM-107/LM-5, Final Flight Plan*, July 1, 1969. Local files: `Apollo11_Flight_Plan.pdf` (353 PDF pages) and `flight_plan.txt`. The text rendition is not an Apollo 13 flight plan. Printed section page numbers below are preferable to text-file line numbers, which change with extraction format.

Source location	What is used
pp. 1-1–1-3	Phase order; orbit insertion; EVA duration; 118.2 min lunar orbit, 47 min night, 72 min MSFN coverage; summary entry / splashdown schedule.
Figure 1-3, p. 1-7	Powered-descent structure; high/low gates motivate the fictional terminal landing decision.
Table 1-1, p. 1-8	CSM ignition times, burn durations, and supplied vector components for TLI, avoidance, LOI, and TEI. PDF page 30.
Table 1-2, p. 1-9	DOI, PDI, ascent, and CSI timing; required velocity changes. PDF page 31.
Detailed timeline, p. 3-69	PDI 102:35:13, burn 11:58, touchdown 102:47:11.
Detailed timeline, p. 3-135	Entry interface 195:03:27, more precise than the summary's 195:03; splashdown EI plus 13:55, giving 195:17:22.

[N1] Supplemental Apollo 13 history. NASA Space Science Data Coordinated Archive, *Apollo 13 Command and Service Module*, catalog ID 1970-029A, "Mission Profile," accessed October 2, 2026. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1970-029A

This source documents the accident at GET 55:54:53, transfer to the LM, an 11.6 m/s return maneuver, the 263.4 s / 262 m/s PC+2 burn, an approximately 15 s later correction, a 22.4 s final correction, LM jettison, and splashdown at GET 142:54:41. Times given in UTC are converted from launch at April 11, 1970, 19:13:00 UTC: PC+2 April 15 at 02:40:39 becomes 079:27:39; LM jettison April 17 at 16:43:02 becomes 141:30:02. The April 16 correction at 04:32 UTC is near 105:19 GET; the game does not present it as an exact ignition timestamp.

What is invented. Emergency and entry battery pools, resource inventories and rates, allowable reserves, simplified thrust / mass models, docking range and speed, terminal landing fuel and altitude, science-station durations, cartridge capacities, manual-navigation components, entry geometry and linear response, checkpoint bands, command reserve, and every error consequence are game design. No supplied Apollo 13 flight plan was available. Historical Apollo 13 event anchors supplement the Apollo 11 plan's flight-planning structure; its rescue path is not an Apollo 11 timeline with names changed.

Limits. No numerical model here predicts a true orbit, real atmospheric heating, scrubber chemistry, or mission survival. Fixed rates and average acceleration are intentional educational approximations. All real missions required substantial ground and onboard computation; this game does not claim they were flown using only a slide rule.